

Epicasting: Neural Networks for Epidemic Forecasting

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Epidemics: A Serious Problem

- Epidemic: a (usually rapid) rise in the prevalence of a disease or condition.
- Flu Patterns:
 - Subtropics and Temperate zones: Seasonal, varying intensity
 - Tropics: Year-round, erratic (250K deaths/year)
- Dengue:
 - Correlates to monsoon season
 - Highly variable outbreaks with global spread (50M – 400M cases/year)
 - Mortality rate is around 2-5% when severe dengue cases are treated.

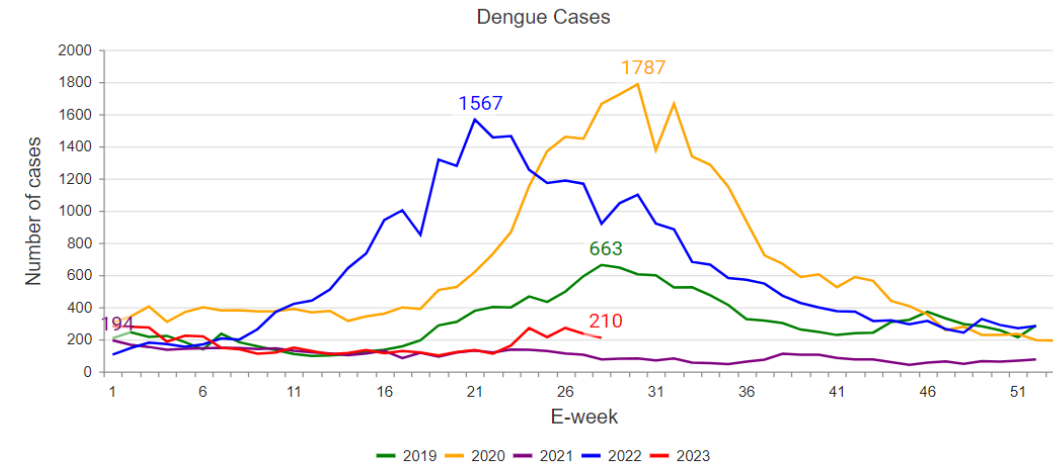


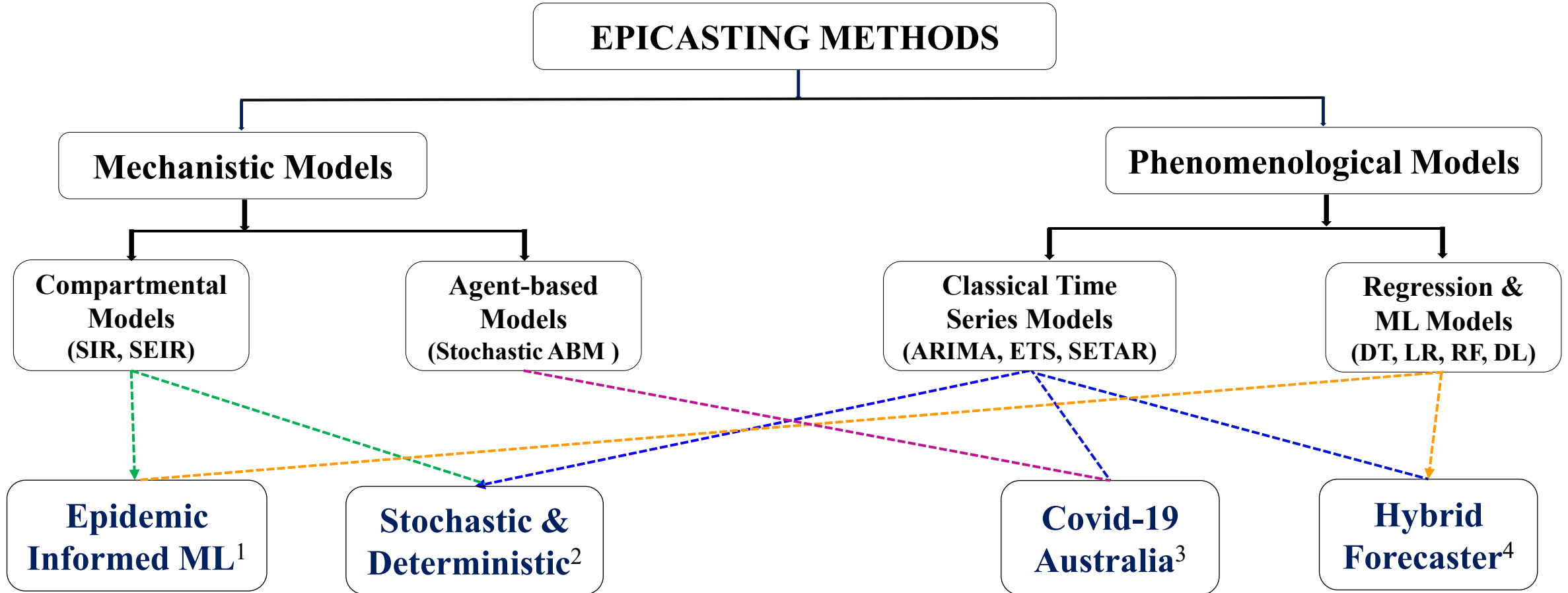
Fig: Weekly dengue incidence cases in Singapore

Benefits of Epicasting

Forecasting is estimating how the sequence of observations (incidence cases) will continue into the future.

- To Governments:
 - Timing and focus of communications (e.g., vaccination campaigns)
 - Antiviral policy
 - Mosquito control, door-to-door campaigns
- To Health Care Providers:
 - Staffing, vacations
 - Elective surgery
 - Equipment pre-positioning
- To Individuals:
 - Protect our families (old and comorbidity)

Epicasting Approaches



¹Rodríguez, Alexander, et al. "Epidemic Forecasting with a Data-Centric Lens." KDD (2022)

²Chakraborty, Tanujit, et al. "Real-time forecasts and risk assessment of novel coronavirus (COVID-19) cases: A data-driven analysis." *Chaos-D* (2020)

³Moss, Robert, et al. "Forecasting COVID-19 activity in Australia to support pandemic response: May to October 2020." *Scientific Reports* (2023)

⁴Chakraborty, Tanujit, et al. "Forecasting dengue epidemics using a hybrid methodology." *Physica A* (2019)

Common Features of Epidemic Data

In general, epidemic datasets are complex and noisy in nature.
They represent the following behavioral characteristics

- ✓ **Non-stationarity** - The statistics of the epidemic time series change over time.
- ✓ **Non-linear** - The process generating the epidemic incidence over time does not follow a linear pattern.
- ✓ **Long-term dependent** - The analysis of the epidemic time series suggests that they possess long memory and the rate of decay of statistical dependence of two points in the series is slower than an exponential decay.
- ✓ **Seasonal** - Another essential characteristic of an epidemic time series is its tendency to repeat its patterns at subsequent intervals.

Monthly dengue incidence (Bangkok)



Ollech and Webel's combined seasonality test – **Monthly seasonality of time series.**

Epidemic Data

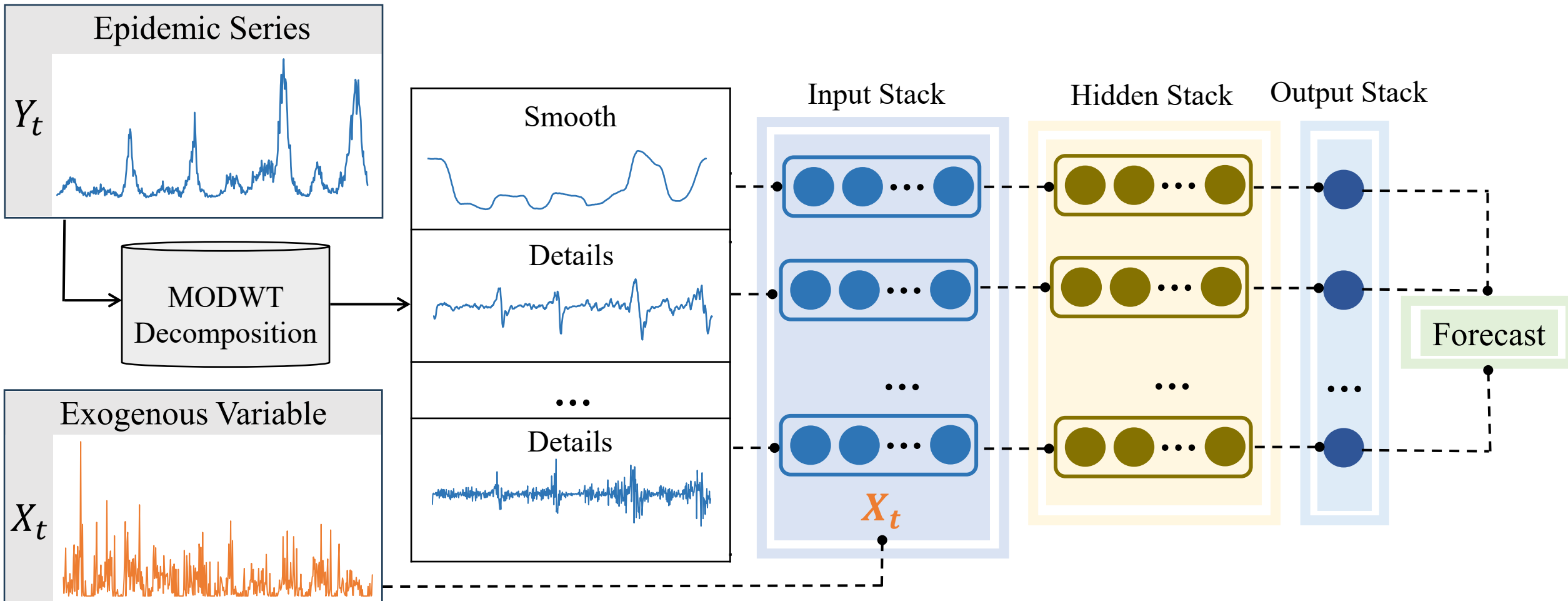
Datasets	Time-span	Frequency	Length	Non-stationary	Nonlinear	LRD
Australia Influenza	1947 – 2015	Weekly	974	✓	✓	✓
Japan Influenza	1998 – 2015	Weekly	964	✗	✓	✓
Mexico Influenza	2000 – 2015	Weekly	830	✓	✓	✓
Ahmedabad Dengue	2005 – 2012	Weekly	424	✓	✓	✓
Bangkok Dengue	2003 – 2017	Monthly	180	✓	✓	✓
Colombia Dengue	2005 – 2016	Weekly	626	✓	✓	✓
Hong Kong Dengue	2002 – 2017	Monthly	192	✓	✗	✓
Iquitos Dengue	2002 – 2013	Weekly	598	✗	✓	✓
Philippines Dengue	2008 – 2016	Monthly	108	✗	✓	✓
San Juan Dengue	1990 – 2013	Weekly	1196	✗	✓	✓
Singapore Dengue	2000 – 2015	Weekly	838	✓	✗	✓
Venezuela Dengue	2002 – 2014	Weekly	660	✓	✗	✓
China Hepatitis B	2010 – 2017	Monthly	92	✓	✓	✓
Colombia Malaria	2005 – 2016	Weekly	626	✓	✗	✓
Venezuela Malaria	2002 – 2014	Weekly	669	✓	✓	✓

Objectives

- 1) Building a hybrid epidemic forecasting model to handle the data irregularities
- 2) State-of-the-art performance
- 3) Understanding the theoretical and computational aspects
- 4) Building software for public use (similar to weather forecasts)
- 5) Making an impact in healthcare

Proposed EWNet Model

- Finally, the one-step ahead forecasts are aggregated to generate the corresponding forecast for the epidemic series Y_t . Multi-step ahead forecasts are obtained recursively.



Proposed EWNet Model

- $J + 1 = \lfloor \log_e(\text{length of training data}) \rfloor$ is the number of wavelet levels (Percival et al., 1997)
- Mathematically, EWNet (p, k) can be written as follows:

$$\hat{Y}_t = \sum_{j=1}^J f_j(D_{j,t}) + f_0(S_{J,t}),$$

where f_i is the one-hidden layered feedforward ARNNx with p lagged inputs and k hidden nodes.

$$f_j(D_{j,t}) = \alpha_{0,j} + \sum_{i=1}^k \beta_{i,j} \phi(\alpha_{i,j} + \beta'_{i,j} \underline{D}_j + \gamma_j x_t)$$

$$f_0(S_{J,t}) = \eta_0 + \sum_{i=1}^k \delta_i \phi(\eta_i + \delta'_i \underline{S}_J + \lambda_j x_t)$$

where $\underline{D}_j, \underline{S}_J$ denotes $p - 1$ lagged observations. $\beta_{i,j}, \eta_i, \delta_i$ are the connection weights of the network, $\beta'_{i,j}, \delta'_i$ are p dimensional weight vectors, and ϕ is the logistic sigmoidal activation function.

- Choice of p and k ?
- Estimator complexity control!

Percival, Donald B., and Harold O. Mofjeld. "Analysis of subtidal coastal sea level fluctuations using wavelets." *JASA* (1997).

Faraway, Julian, and Chris Chatfield. "Time series forecasting with neural networks: a comparative study using the air line data." *JRSS-C* (1998)

Stable learning in EWNet

Choice of p and k

- We select p based on the minimum error metric on the validation set (V)

$$p = \operatorname{argmin}_p \frac{1}{|V|} \sum_{t \in V} \frac{2|\hat{Y}_t - Y_t|}{|\hat{Y}_t| + |Y_t|} * 100\%$$

- From the boundary of stable learning in hidden and output neurons, we introduce a balancing equation as:

$$\alpha \eta \left(\frac{p}{k} \right) = \eta k,$$

where η is the learning rate and α is a consistency constant with $1 \leq \alpha \leq p$. Thus, we initially choose $k = \sqrt{\alpha p} \Rightarrow k \in (\sqrt{p}, p)$ and using the AM-GM inequality we finally select $k = \left\lfloor \frac{p+1}{2} \right\rfloor$.

Practical Significance

- The number of hidden nodes in the EWNet model is set to a fixed value depending on the number of lagged inputs.
- Due to this, the **running time of the EWNet model is minimal** as compared to unstable neural networks in which the number of hidden nodes either becomes too large or too small.

Empirical Risk Minimization in EWNNet

Proposition

Let the autoregressive neural network (of order k) is applied to the original data Y_t by minimizing the risk $\mathfrak{R}_{Emp}$ and EWNNet fits the ensemble model on the transformed data by minimizing the empirical risk $\mathfrak{R}_{Emp}^W$, then we have

$$\min \mathfrak{R}_{Emp}^W \leq \min \mathfrak{R}_{Emp}$$

$$\mathfrak{R}_{Emp}^W = \frac{1}{N} \sum_{i=1}^N \left[(S_{J,i} - \hat{S}_{J,i}) + \sum_{j=1}^J (D_{j,i} - \hat{D}_{j,i}) \right]^2$$

$$\mathfrak{R}_{Emp} = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_{ARNN_t})^2$$

Practical Significance

- The above result shows the **robustness** of the wavelet decomposed approach in EWNNet from an ERM perspective.

Proposed EWNet Model

- h -step ahead forecasts of Y_t from EWNet(p, k) model based on N historical observations and exogenous variables can be expressed as:

$$\hat{Y}_{N+h} = \sum_{j=1}^J \hat{D}_{j,N+h} + \hat{S}_{J,N+h}$$

where, $\hat{D}_{j,N+h}$, and $\hat{S}_{J,N+h}$ are the h -step forecasts for the details and smooth coefficients.

From a statistical point of view, we study the associated Markov Chain for the EWNet process.

Asymptotic stationarity



This theoretical result guarantees that the EWNet model cannot show ‘explosive’ behavior or growing variance over time. Helps in model specification and estimation during simulation study.

Uncertainty Quantification in EWNet

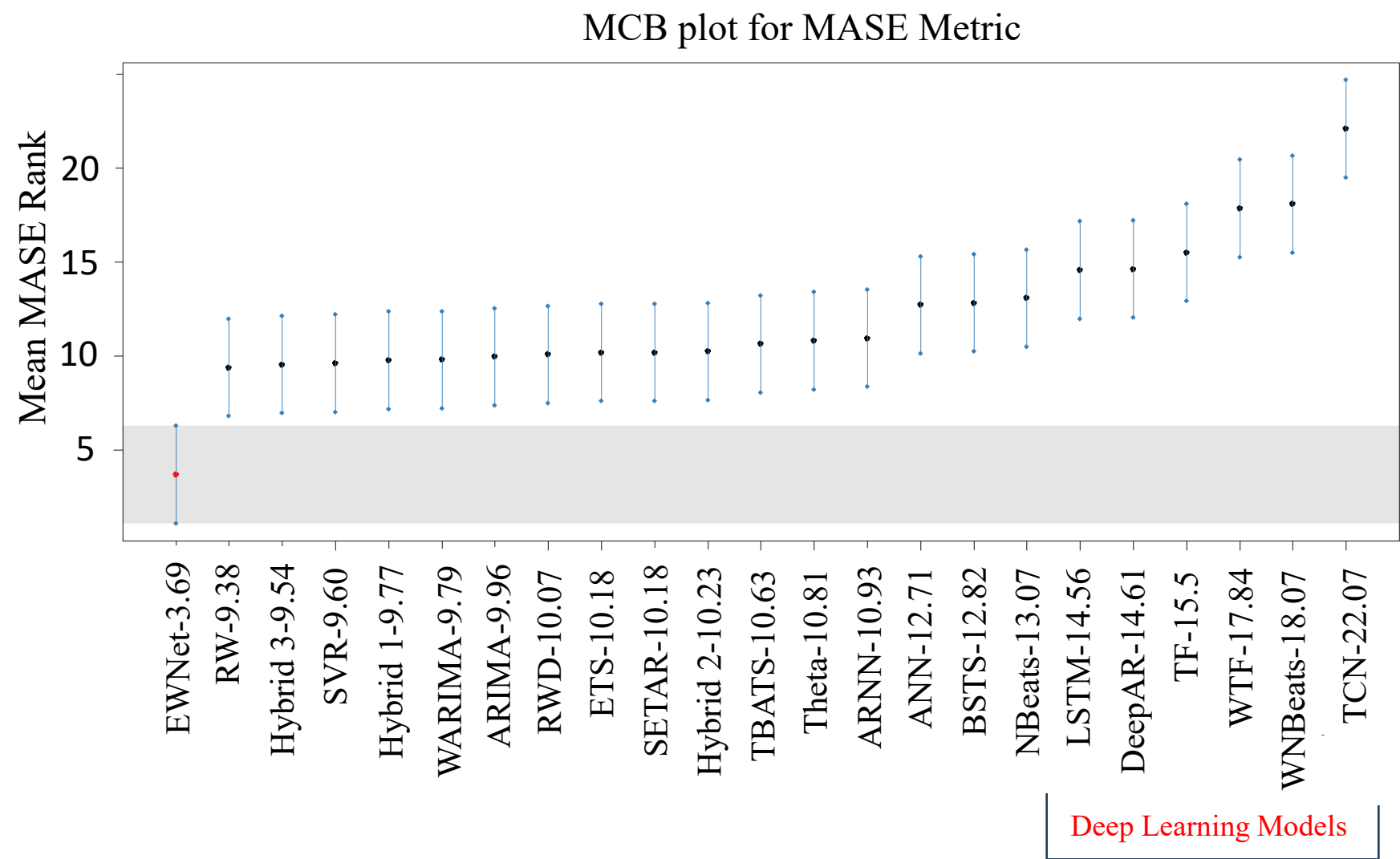
- Going **beyond point forecast** by providing valid prediction intervals for individual forecasts.
- EWNet leverages the distribution-free **Conformal Prediction** (CP) technique for estimating uncertainty in predictions.
- In CP method, **conformal score** (S_t) is calculated based on the absolute residuals standardized with the conditional absolute deviation (uncertainty model: $\hat{Y}_t^U$)

$$S_t = \frac{|Y_t - \hat{Y}_t|}{\hat{Y}_t^U}$$

- The CP interval for prediction $\hat{Y}_t$ at time step t can be computed using the **conformal quantiles** ($\hat{\mathbb{Q}}_t$) as:

$$\mathbb{C}(\hat{Y}_t) = [\hat{Y}_t - \hat{\mathbb{Q}}_t \hat{Y}_t^U, \hat{Y}_t + \hat{\mathbb{Q}}_t \hat{Y}_t^U]$$

MCB test results



Schematic visualization of the multiple comparisons with the best (MCB) test.

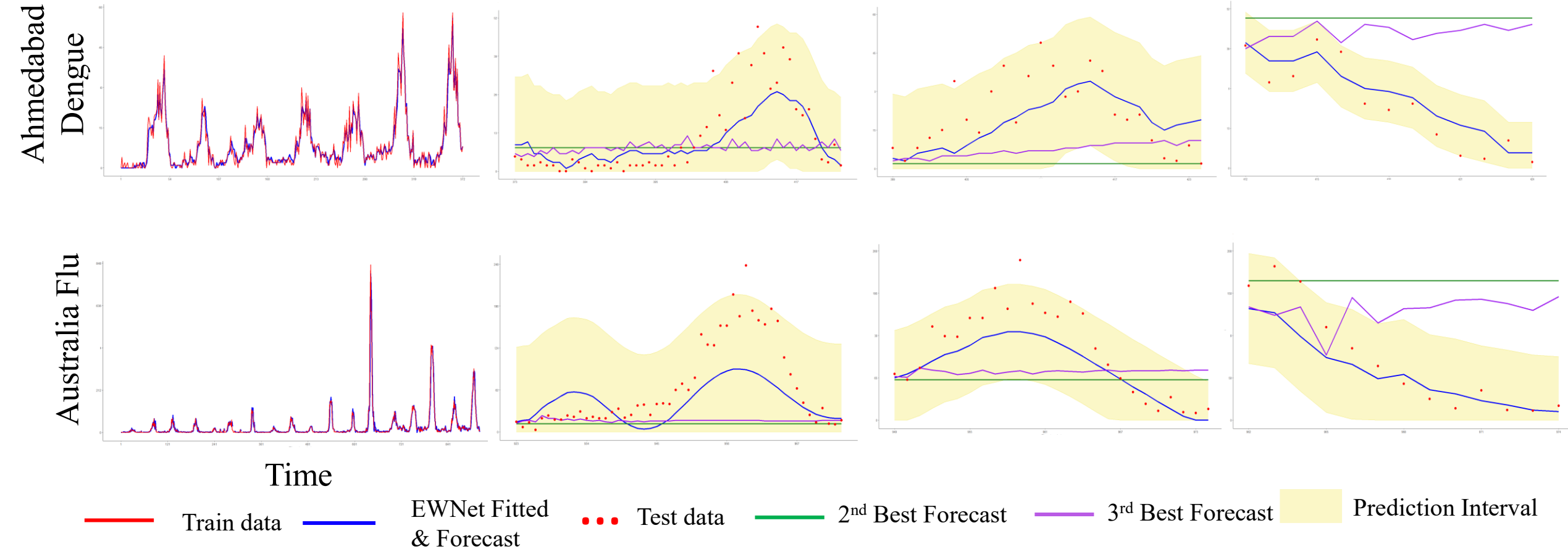
Actual vs. Forecast Visualization

Training data and fitted values

Long-term forecast

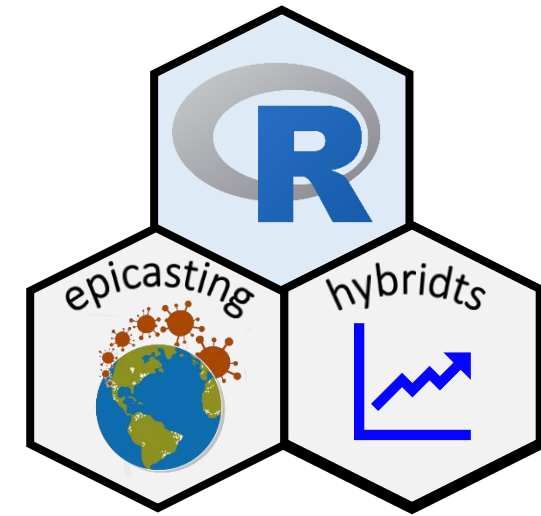
Medium-term forecast

Short-term forecast



Remarks

- Suitable for non-stationary, long-range dependence, and non-linear forecasting problems with limited historical data.
- Theoretical properties ([learning stability](#), [geometric ergodicity](#), and [asymptotic stationarity](#)) ensure the stability of model output.
- Simple and easily interpretable model, fast in implementation due to pre-defined architecture.
- [PharamaACE](#) adopted EWNet as a digital tool for predicting epidemic outbreaks in SE Asia.
- [Extension on Progress](#): Data + Drivers (geographical scale, vegetation, and other attributes that impact epidemics) + ODEs + NN ~ Model for opendengue data



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Publication



Neural Networks

Volume 165, August 2023, Pages 185-212

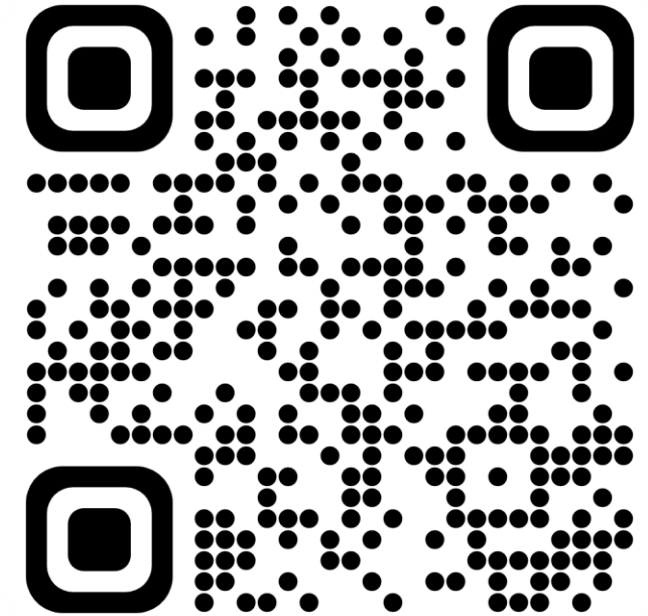


Epicasting: An Ensemble Wavelet Neural Network for forecasting epidemics

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